

Jonathan Setiawan

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SUMMARY / PROFESSIONAL OBJECTIVE

Robotics and AI researcher building perception and learning systems for physical robots with interest in robot perception modalities, SciML, and physics in robotic systems

EDUCATION

Ritsumeikan University (立命館大学) **Osaka, Japan**
College of Information Systems Science and Engineering *Graduation Date: Mar 2028*
Awards: Murai SIIIX Scholarship (2025 & 2026), JASSO Monbukagakusho Honors Scholarship (2024)
Certifications: IELTS 8.0 (2022) *GPA: 4.45/5.0*

TECH STACK

Python, C, C++, Java, JavaScript/TypeScript, MySQL, ROS2 (Humble), Gazebo, OpenCV, PyTorch, TensorFlow, SLAM, Docker, Git, React, CMake, LaTeX, Linux, Go, Isaac Sim, MuJoCo, Rust

RESEARCH & LEADERSHIP EXPERIENCE

FPTジャパン株式会社 **Tokyo, Japan**
Physical AI Team Robot Learning Engineer *May 2026 - Aug 2026*

- Trained and evaluated imitation-learning policies (ACT, Diffusion Policy variants, SmolVLA) on the LeRobot framework, including MolmoAct for grounded trajectory planning from 2.5D visual input
- Built simulation environments for robot arm manipulation in MuJoCo, Gazebo, and NVIDIA Isaac Sim, and ran sim-to-real transfer of learned policies onto physical hardware

iCOM技研株式会社 **Hyogo, Japan**
Robotics ML Engineer *Jul 2025 - Present*

- Developed the perception stack for robotic parcel picking: trained and evaluated RGB-D fusion models and point cloud segmentation models to detect suctionable top surfaces on unknown parcels
- Developed a pipeline for class-agnostic segmentation with synthetic RGB-D training using BlenderProc domain randomization to generalize to unseen parcel types
- Architected the end-to-end palletizing/depalletizing system for UR5e, UR10e, and UR20 robots

Ritsumeikan University (立命館大学) **Osaka, Japan**
RiOne (Robocup @Home League) Robotics Club Computer Vision Team Lead *Jul 2024 - Present*

- Architected a multi-node ROS 2 Humble perception stack and sensor-fused navigation on TurtleBot3 WafflePi with face re-identification and temporal tracking for persistent HRI and autonomous obstacle avoidance
- Engineered a self-improving object recognition pipeline and achieved 3rd Place for Open Challenge and Best Poster Award in RoboCup Japan Open 2026 (Bridge Division)

PROJECT EXPERIENCE

SciBERT-CTFT: Scientific Abstract Classifier via Contrastive Learning Fine-tuning
Artificial Intelligence Course Project *Dec 2025 - Jan 2026*

- Fine-tuned a SciBERT model using the SetFit framework to classify scientific literature with 99.61% accuracy, implementing a novel context injection data augmentation technique using TF-IDF maps

EFT-VPR: Event Forecasting Transformer for Visual Place Recognition
Personal Project *Feb 2026 - Mar 2026*

- Engineered an event-based localization stack utilizing a 3-layer SNN encoder and an 8-head autoregressive Transformer with InfoNCE contrastive learning, improving Recall@1 by 13.8% in a 65,992-frame dataset
- Designed a VPR pipeline leveraging FAISS (IndexFlatIP/GPU) for top-K cosine retrieval and dropout-robust forecasting, reducing median localization error by 27.8% (81.9m to 59.1m)

Foveated Teleoperation Task-based Study
PBL5 project *Apr 2026 - July 2026*

- Architected a real-time foveated teleoperation pipeline streaming headset eye-tracking data via TCP/UDP to an encoding server, and onward to a rover mini-PC with ESP32 motor control
- Engineered a 90 Hz telemetry logger and implemented gaze-signal filtering with velocity-based fixation classification for low-latency robotic control